Data Manipulation

→ df.describe() → statistical summary of numerical columns.

→ df.dtypes → data type of each column

① Handling missing values

→ To check

>>> df.isnull() → returns which values are null

>>> df.isnull().any(axis=1) → by default = 0.

or.

>>> df.isnull().sum()

Ans: df1=df.fillna(0) → fills all missing values with 0

or df['filled'] = df['target'].fillna(df['target'].mean())

→ creates a new column, a copy of target column, with empty values filled with mean of the column.

Vimp

Renaming columns

~~df.rem~~

~~df = df.rename~~

updates the column name 'Date' with 'Sales Date'

```
df = df.rename(columns = {'Date': 'Sales Date'})
```

df.rename()

To be replaced

New replacement name

Changing Data types

filling missing values with mean

```
df['New-val'] = df['Val'].fillna(df['Val'].mean())
                          .astype(int)
```

new column

original col.

converting type to int.

Applying a function to a column

```
df['New'] = df['Val'].apply(lambda x:x*2)
```

New col

df.apply(function name)

→ creates a new column 'New' with values that are 2 * x for x in 'Values'.

Data Aggregating & Grouping

```
grouped_mean = df.groupby('Product')['Value'].mean()
```

df.groupby(col)

categorical variable

gives the mean() of the grouped values.

for grouping based on 2 ~~cols~~ categories

```
grouped_sum = df.groupby(['A', 'B']).['Value'].sum()
```

new variable

list of columns

gives the sum of values for all groups

Applying multiple functions?

```
group_agg = df.groupby('A').['Value'].agg(['mean', 'sum', 'count'])
```

['Value'].agg(funs list)

grouping df based on 'A'

list of functions to be applied

Merging 2 Dfs

```
new = pd.merge(df1, df2, on = 'Key', how = 'inner')
```

- new → saving the merged df
- df1, df2 → dataframes to be merged
- on = 'Key' → common parameter
- how = 'inner' → type of join
  - left
  - right
  - inner
  - outer

- → left join → left df's keys
- → right → right df's keys
- → outer → all keys
- → inner → common keys